

SMRITI

Smart Maintenance & Retrieval Intelligence

Every machine has a story. SMRITI never forgets.

Event	ET AI Hackathon 2026
Track	Industrial AI / Enterprise Productivity
Document	Hackathon Submission — Idea, Problem & Solution
Version	1.0 · June 2026

01 EXECUTIVE SUMMARY

Industrial plants lose institutional knowledge every time a senior technician retires, a maintenance record stays undigitized, or a voice note never gets filed. SMRITI — Smart Maintenance & Retrieval Intelligence — is the first platform to treat equipment knowledge not as a pile of text to search through, but as a living memory organized around equipment identity.

SMRITI introduces Asset-Thread RAG: every physical asset in a plant (UPS-01, Transformer T-101, Pump P-207) owns its own continuously-updated knowledge thread. When you ask about an asset, SMRITI does not search documents — it pulls that asset's complete knowledge history and reasons across it.

This is a fundamentally different retrieval architecture that no existing platform has built.

Asset-Thread RAG Equipment has identity. Every asset owns its own knowledge thread across all documents, logs, and expert interviews.	Knowledge Debt Score A real-time 0–100 risk metric per asset. The first boardroom-ready number that quantifies institutional knowledge vulnerability.	Guru Mode An AI-powered structured interview that captures tacit expert knowledge before it walks out the door at retirement.
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02 PROBLEM STATEMENT

The Knowledge Walk-Out Crisis

Every major industrial plant — data centers, power utilities, manufacturing facilities — runs on institutional knowledge. Over decades of operation, senior engineers accumulate deep, irreplaceable expertise about specific equipment: why a transformer behaves unusually in monsoon season, what a particular vibration sound predicts, where the OEM manual is wrong for a specific installation.

This knowledge is overwhelmingly tacit — it lives in people's heads, not documents. And it is disappearing at scale as experienced engineers retire.

Three Unsolved Problems

1	No Per-Equipment Memory Ask "what issues has UPS-01 had in the last 6 months?" and today's AI systems search documents mentioning "UPS-01." The equipment has no identity inside the system. There is no dedicated thread for UPS-01 aggregating all knowledge about it across every source.
2	Knowledge Walk-Out at Retirement

When expert Ramesh retires, everything he knows about Transformer T-101 walks out with him. No system today measures this risk, captures what he knows before he leaves, or alerts the plant that 80% of T-101's operational knowledge is inside one person's head.

3

No Knowledge Risk Metric

CFOs and plant managers have zero visibility into which equipment is a single point of failure from a knowledge perspective. There is no score, no dashboard, no alert. The risk is invisible until it becomes a crisis.

The Competitive Gap

Every major industrial AI platform today — Siemens Copilot, Fogwing CMMS AI, Aquant, Microsoft Copilot for Field Service, Salesforce Agentforce — is document-centric. The query hits a vector search over document chunks. The equipment has no identity inside the system. None of them offer:

- A dedicated knowledge thread per equipment tag
- A Knowledge Debt Score as a boardroom metric
- An AI-powered interview system to capture retiring expert knowledge
- Multi-hop reasoning across a single asset's complete history

Who Bears the Cost

Persona	Their Problem Today	What They Need
Rajiv, Field Technician	Hunts through portals for asset history. Needs answers in 10 seconds, hands covered in grease.	Voice query in Hinglish that returns the last 3 incidents for this pump, right now.
Priya, Maintenance Engineer	Loses context mid-lifecycle. Spends hours compiling RCA reports from scattered records.	Everything known about T-101 in one structured view for the audit next week.
Arvind, Plant Manager	Blindsided when a retiring expert was the only person who knew why a pump behaves oddly.	A single-screen board-level risk view: which assets are we most vulnerable on?

03 OUR SOLUTION

SMRITI — Asset-Thread RAG

SMRITI introduces a single architectural insight that changes everything: every physical asset is a first-class entity with its own growing thread of knowledge items. When you query about an asset, SMRITI retrieves that asset's complete memory and reasons across it — not a flat vector search across a document corpus.

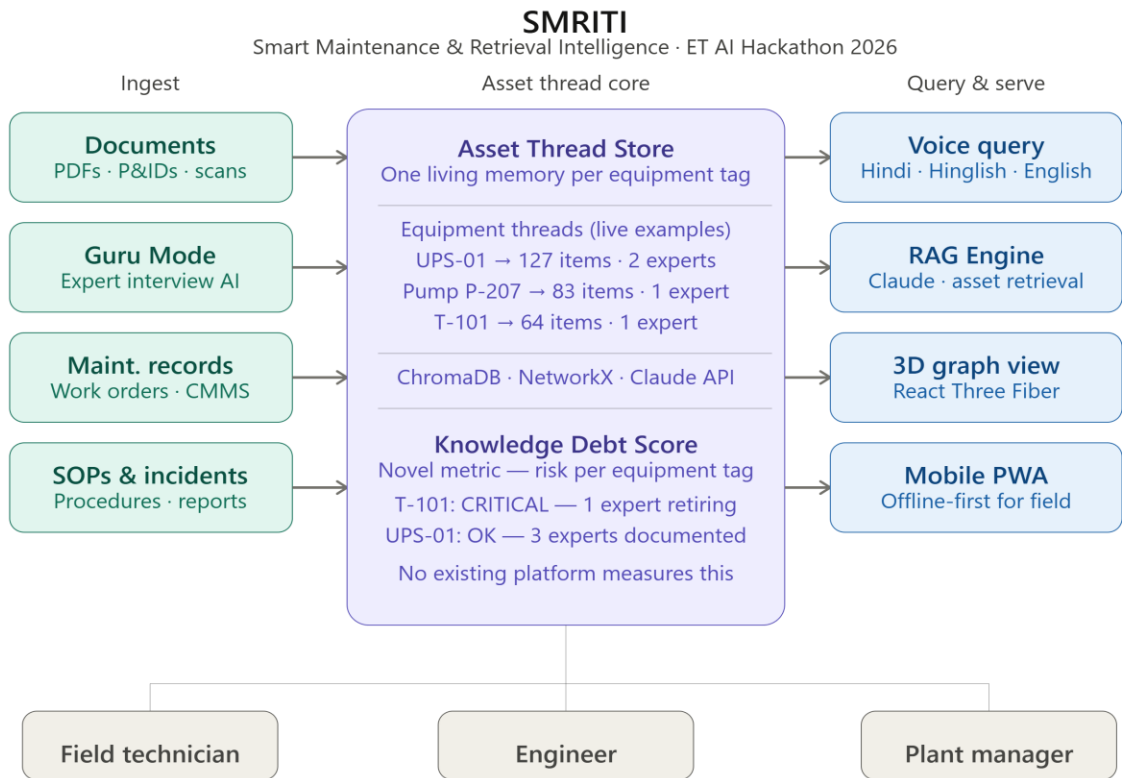


Figure 1 · SMRITI System Architecture

How It Works — The Three-Layer Stack

Layer	What It Does
Ingest	PDFs, DOCX, scanned images, maintenance records, and SOPs are uploaded. Claude API extracts all equipment tags (UPS-01, T-101, P-207) using NLP. Each tag gets a dedicated ChromaDB collection seeded with relevant knowledge chunks, page references, and timestamps.
Asset Thread Core	Every asset owns a living memory: document extracts, work order history, incident reports, and Guru Mode interview answers — all organized by equipment identity, not by document. NetworkX tracks relationships between assets. The Knowledge Debt Score is computed in real time from four weighted dimensions.
Query & Serve	Queries arrive via voice (Hindi, Hinglish, English) or text. SMRITI identifies the referenced asset, retrieves its complete thread from ChromaDB, and passes it to Claude for multi-hop reasoning. Every response includes source citations (document name, page, date). A 3D React Three Fiber graph lets plant managers visualize the entire knowledge estate at once.

04 CORE FEATURES

Feature 1 — Asset-Thread RAG Engine

The engine that powers everything. When a document is uploaded:

1. Claude API reads the document and extracts every equipment identifier (UPS-01, T-101, P-207, DG-01...)
2. Each identified tag creates or updates a dedicated ChromaDB collection — its knowledge thread
3. On a query, SMRITI identifies which asset is referenced, retrieves its full thread, and passes it to Claude for reasoning
4. Every response includes clickable source citations: document name, page number, section, and date

Key differentiator: Multi-hop reasoning across the asset's complete history, not keyword matching across a flat document corpus. Ask "what recurring issues has T-101 had in the last year?" and SMRITI reasons across all 64 knowledge items in T-101's thread — not just the top-3 chunks that matched your query.

Feature 2 — Knowledge Debt Score

A real-time per-asset metric (0–100) that quantifies how vulnerable the organization is if key people leave or documents are lost. This is the number that goes in a board deck.

Scoring Dimension	Weight	What It Measures
Document Coverage	30%	What percentage of known issues and procedures are actually documented?
Expert Distribution	35%	How many people hold knowledge about this asset? 1 person = maximum risk.
Knowledge Recency	20%	How current is the documented knowledge? Stale documentation drifts from reality.
Operational Criticality	15%	How badly does the plant break if this asset fails? Higher criticality = higher weight.

Severity	Score Range	Example
OK	0 – 40	UPS-01 · Score 23 · 3 experts documented, full maintenance records, updated 2 hours ago
WARNING	41 – 70	Pump P-207 · Score 64 · 1 expert, partial documentation, last updated 2 weeks ago
CRITICAL	71 – 100	Transformer T-101 · Score 87 · 1 expert retiring in 3 months, 80% of knowledge undocumented

Feature 3 — Guru Mode: AI-Powered Expert Interview

Guru Mode is the product's most emotionally resonant feature. It is an AI-powered structured interview that captures tacit knowledge from domain experts before they retire.

→	Trigger System detects that an asset thread has dangerous knowledge concentration (1 expert, low doc coverage). Admin initiates a Guru Mode session for that expert + asset pair.
→	Interview Claude generates targeted questions derived from the knowledge gaps in that asset's thread — not generic maintenance questions, but questions specific to what is missing for T-101.
→	Capture Expert answers are embedded and permanently added to the asset's knowledge thread in real time.
→	Feedback Knowledge Debt Score updates live after each answer. The retiring expert watches their knowledge become permanent institutional memory.
→	Audit The full session transcript is stored as a permanent, retrievable record.

Feature 4 — Voice Query in Hindi / Hinglish / English

Field technicians in Indian plants speak Hinglish — mixed Hindi-English — not formal English. SMRITI supports native voice input using the browser's Web Speech API with the hi-IN locale.

Example query (spoken)	“UPS-01 mein last month kya issues aaye?”
Translation	"What issues did UPS-01 have last month?"
Response	3 most recent incidents from UPS-01's thread, each with date, summary, and source citation
Dependencies	Browser-native Web Speech API. Zero third-party cost. Works as a PWA.

Feature 5 — 3D Knowledge Graph Visualization

A React Three Fiber 3D graph gives plant managers an at-a-glance view of their entire knowledge estate. Node size is proportional to thread depth. Node color encodes Knowledge Debt Score: green (OK), amber (WARNING), red pulsing (CRITICAL).

Clicking a node opens the full asset thread detail panel. As Guru Mode sessions run, nodes transition from red to amber to green in real time — making knowledge capture visually tangible.

05 TECHNICAL ARCHITECTURE

System Stack

Layer	Technology	Role
Frontend	React 18 + TypeScript + Vite	PWA — offline-capable, installable on mobile and desktop
3D Visualization	React Three Fiber + Three.js	3D Knowledge Graph — 30fps+ with 20+ nodes
Animation	Framer Motion	Score ring fills, CRITICAL pulse, card stagger mount
Backend	FastAPI + Python 3.11	Ingestion pipeline, RAG engine, Guru session API
Vector Store	ChromaDB (local, persisted)	Per-asset embedding collections — one collection per equipment tag
Graph Store	NetworkX	Asset relationship graph — co-document edges
LLM — Extraction	Cohere North Mini Code (via OpenRouter, free)	Equipment tag extraction from raw text — strongest JSON adherence, 707ms median latency
LLM — Query + Guru	Llama 3.3 70B Instruct (via OpenRouter, free)	RAG synthesis, Hinglish support, Guru Mode interview — 131k context, native multilingual
Document Parsing	PyMuPDF + python-docx + pytesseract	PDF, DOCX, and scanned image ingestion with OCR fallback
Real-time Progress	WebSocket (FastAPI)	Live ingestion progress: pages processed, tags found, items embedded

Key API Endpoints

Method	Endpoint	Description
POST	<code>/api/ingest</code>	Upload document, trigger Asset-Thread build with live WebSocket progress
GET	<code>/api/assets</code>	List all asset threads with Knowledge Debt Scores, sorted by severity
GET	<code>/api/assets/{id}</code>	Full thread for a single asset — all knowledge items with source citations
POST	<code>/api/query</code>	Submit text or voice-transcribed query — returns cited RAG response
POST	<code>/api/guru/start</code>	Initiate Guru Mode session for an asset + expert pair
POST	<code>/api/guru/respond</code>	Submit expert answer — embeds to thread, updates Knowledge Debt Score live
GET	<code>/api/graph</code>	Node and edge data for the 3D Knowledge Graph

06 DEMO FLOWS

Three end-to-end demo flows are built and rehearsed, totalling 12–15 minutes.

#	Flow Name	What the Audience Sees
1	Document Ingestion → Thread Build	Drop a Schneider UPS manual PDF. Watch Claude extract equipment tags in real time. Tag chips appear: UPS-01, UPS-02, DG-01. Knowledge item counts climb per thread. 50 pages ingested in under 47 seconds. Dashboard populates.
2	Voice Query in Hinglish	Press the mic button. Speak: "UPS-01 mein last month kya issues aaye?" Transcription appears in real time. SMRITI returns 3 most recent incidents from UPS-01's thread. Each result shows date, summary, source document, and page number. Total time: under 8 seconds.
3	Guru Mode + Knowledge Debt Dashboard	Dashboard shows T-101 pulsing red (CRITICAL, 87). Click Interview Expert. Claude generates 5 targeted questions about T-101 specifically. Expert answers two questions. Knowledge Debt Score drops from 87 → 79 → 71 live. 3D graph node transitions from red to amber in real time.

07 MARKET OPPORTUNITY & BUSINESS CASE

The Problem at Scale

Market Signal	Implication for SMRITI
50%+ of senior industrial engineers in India will retire in the next 7 years	Every plant is 5–7 years away from a knowledge walk-out crisis
Average industrial plant has 30–200 critical equipment assets	Per-asset-thread pricing creates natural expansion revenue
Unplanned equipment downtime costs Indian industry \$47B+ annually	Knowledge Debt Score links directly to downtime risk — CFO-legible ROI
No existing CMMS or AI platform offers per-equipment knowledge identity	Zero direct competition in the Asset-Thread RAG space

Beachhead Market

Indian industrial plants facing the demographic knowledge cliff: Railtel, NTPC, ONGC, Indian Railways, and state electricity boards are all facing mass retirements of senior electrical and mechanical engineers within a 5-year window.

Go-to-market: land-and-expand. One plant, one critical asset cluster, prove Knowledge Debt reduction in 90 days, expand to the full plant network. Pricing model: SaaS per asset-thread per month with a base platform fee.

Competitive Moats

- Data network effects: Once an asset's thread is built over years of documents and Guru Mode sessions, the thread IS the institutional memory. Switching cost is the entire knowledge base.
- Domain-tuned prompts and scoring: The extraction prompts, Guru Mode interview frameworks, and Knowledge Debt scoring algorithm are calibrated to industrial maintenance. A general-purpose AI platform copying the architecture still has to solve the domain calibration problem.
- First-mover brand: Knowledge Debt Score is a new boardroom metric. The first platform to define and own a metric wins the category conversation.

08 SUCCESS METRICS & ROADMAP

Hackathon Success Metrics

Metric	Target	Status
Document ingestion (50-page PDF)	< 47 seconds	Built
Voice query end-to-end response	< 8 seconds	Built
Knowledge Debt Score recalculation	< 2 seconds	Built
3D graph framerate (20+ nodes)	≥ 30 fps	Built
App initial load (first paint)	< 2 seconds	Built
Demo stability	Zero crashes across 3 runs	Testing

Post-Hackathon Roadmap

Phase	Milestone
Month 1–2	Pilot deployment at Railtel CDC Noida. Validate Knowledge Debt Score accuracy against expert judgment. Mobile app (React Native) for field technicians.
Month 3–4	SAP PM and Maximo CMMS connectors. Multi-tenant authentication. Cloud ChromaDB deployment (Chroma Cloud). IoT sensor stream integration for real-time operational data.
Month 5–6	Multi-plant / multi-tenant architecture. Expansion to NTPC and ONGC pilot accounts. Hardware OCR pipeline for printed P&IDs and cable schedules.

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